

Project 1: Autonomous Multi-Agent Market Intelligence & Outreach Pipeline

Goal: Build an AI platform that autonomously researches companies, identifies opportunities, writes personalized outreach, validates quality, and prepares campaigns while allowing human approval before sending.

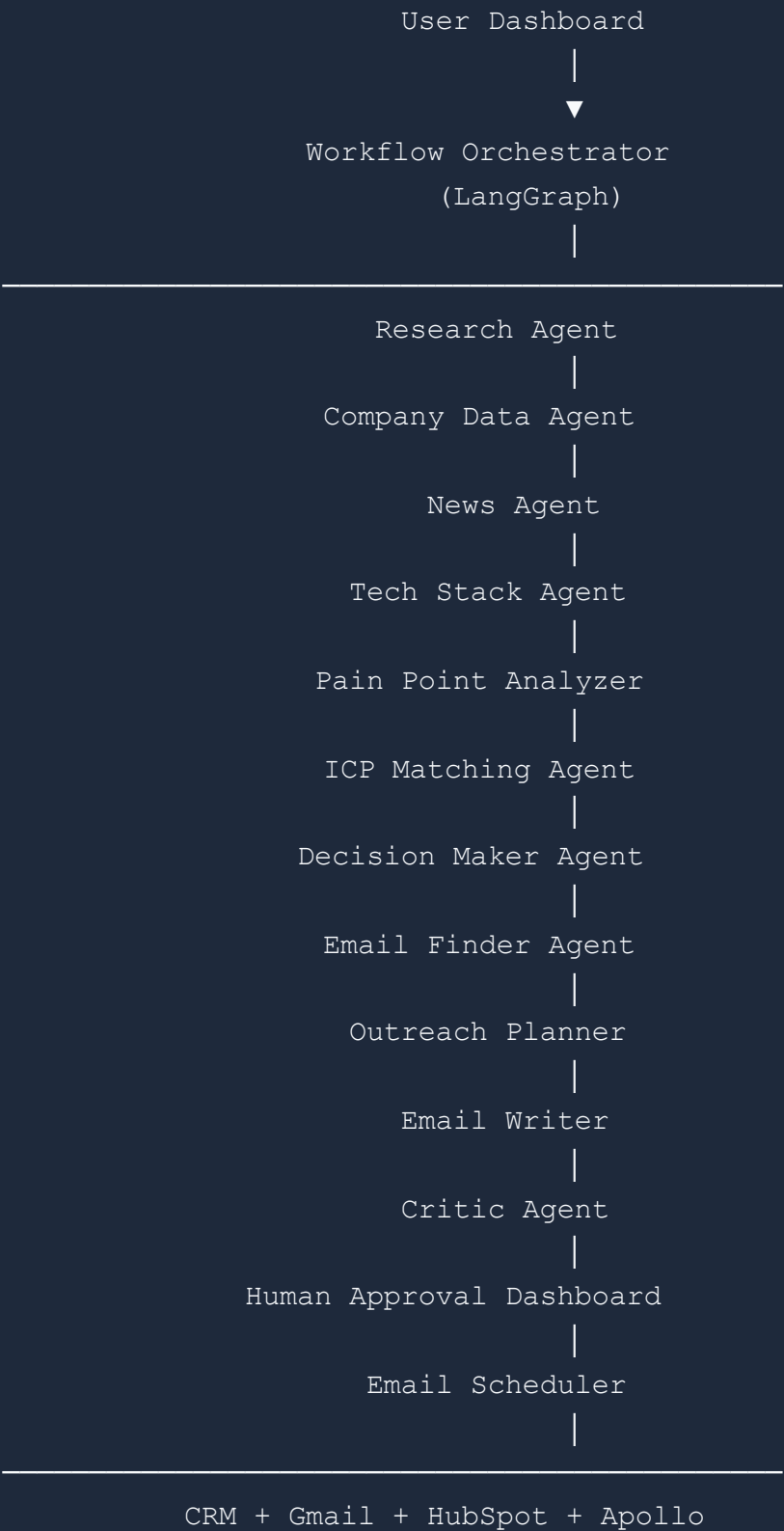
Think of it as an AI Sales Development Representative (AI SDR).

1. Business Problem

Suppose a software agency wants to sell AI automation services.

Manual Process Today	AI System Workflow
• Search companies • Read company websites • Read LinkedIn & recent news • Identify pain points • Find decision makers • Find email addresses • Write personalized cold emails • Schedule follow-ups	The AI system completely automates this pipeline through a multi-agent orchestration layer, gathering data, synthesizing context, and drafting campaigns in seconds.
Time: 20–40 minutes per company.	Time: 2–3 minutes per company.

2. System Architecture



3. Complete Tech Stack

Backend & AI

- **Backend:** FastAPI
- **Framework:** LangGraph
- **LLMs:** OpenAI GPT-5 / Claude / Gemini

Search & APIs

- Tavily
- Serper
- Firecrawl
- Jina AI Reader

Databases & Queue

- **SQL:** PostgreSQL
- **Cache/Queue:** Redis, Celery
- **Vector:** Chroma

Frontend & Ops

- **Frontend:** Next.js, React, Tailwind
- **Monitoring:** LangSmith, OpenTelemetry, Prometheus, Grafana

4. Every Agent (Detailed Breakdown)

Agent 1: Company Research

Input: OpenAI

Tasks (Search): Website, About page, Products, Pricing, Customers, Funding, Careers, Press Releases, Blog, GitHub.

Output Format:

```
{
  company_name,
  industry,
  employee_count,
  funding,
  technologies,
  products,
  growth_stage
}
```

Agent 2: Tech Stack Detector

Uses: BuiltWith, Wappalyzer, Firecrawl.

Detects: React, AWS, Azure, Stripe, HubSpot, Salesforce, Zendesk, Shopify, etc.

Output:

- Current stack
- Potential integrations
- Weaknesses

Agent 3: News Research

Reads: Recent news, Funding, Hiring, Acquisitions, Launches, Blog posts, Press releases.

Output Example:

- "Recent Series B"
- "Hiring AI Engineers"
- "Launching Healthcare Product"
- "Expanding Europe"

Agent 4: Pain Point Analyzer

Reads: Website, Job descriptions, News, Customer reviews.

Finds:

- Possible bottlenecks
- Hiring challenges
- Customer complaints
- Scaling issues
- AI opportunities

Agent 5: ICP Matching

Checks: Does this company match our target market?

Metrics Evaluated: Revenue, Employees, Industry, Uses CRM, Has Engineering Team, Growing rapidly, AI adoption score.

Returns: `Qualified` or `Rejected`

Agent 6: Decision Maker Finder

Finds: CEO, CTO, Head of Sales, VP Engineering, Director Operations.

Uses: Apollo, Hunter, LinkedIn, People Data Labs.

Agent 7: Email Finder

Validates:

- Email syntax & Domain
- Deliverability
- Confidence score

Agent 8: Outreach Planner

Creates Strategy Sequence:

- Email 1
- Follow Up
- Case Study
- Demo Offer
- LinkedIn Message

Agent 9: Email Writer

Produces: Subject, Body, CTA, Personalization.

Example output snippets:

- "Saw your recent funding."
- "Many SaaS companies at your stage struggle with..."
- "We recently built..."
- "Would love to show..."

Agent 10: Critic Agent

Evaluates: Hallucination, Spam score, Grammar, Personalization, "AI sounding" tone, Missing variables, CTA quality.

Logic: If score `<90` → Rewrite.

Agent 11: Human Approval Dashboard

Actions:

- Approve
- Reject
- Edit
- Regenerate

Agent 12: Campaign Scheduler

Schedules:

- Email 1
- Follow-up
- Reminder
- CRM update

5. LangGraph Flow



6. Memory System

Type	Storage Tech	What it Stores
Short-term	Redis	Current workflow, Conversation, Errors
Long-term	Postgres	Past outreach, Companies, Campaign history, User preferences
Semantic Memory	Vector DB (Chroma)	Research context, Previous conversations, Similar companies

7. Tool Calling

Each agent is equipped with specific tools to execute its function.

Research Agent Tools

- Google Search
- Firecrawl
- Wikipedia
- SEC filings
- Crunchbase

Email Agent Tools

- SMTP Protocol
- Gmail API
- HubSpot API

8 & 9. Dashboard & Production Features

Dashboard Capabilities

Shows:

- Research overview & Agent thinking
- Token usage & Cost
- Email previews
- Approval queue
- Live workflow graph

Production Features

Retry & Checkpoint

Caching & Streaming

Authentication (RBAC)

API Rate Limiting

Audit Logs

Docker & CI/CD

Monitoring & Tracing

Alerts

10. Portfolio Impact

Demonstrates expertise in:

✓ Multi-Agent Systems

✓ LangGraph

✓ Tool Calling

✓ Long-Term Memory

✓ Production Backend

✓ Human-in-the-loop

✓ Workflow Design

✓ Evaluation

✓ Observability

✓ Cost Optimization